

IIT Foundation Module for Class VIII Math

- Theme-1: Pure Arithmetic
 - Rational & Irrational numbers

- Theme-2: Commercial Mathematics
 - Compound Interest

- Theme-3: Algebra
 - Expansions
 - Factorisation
 - Simultaneous Linear Equations in two variables (With numerical coefficients only)
 - Indices/Exponents
 - Logarithms

- Theme-4: Geometry
 - Triangles
 - Congruency: four cases: SSS, SAS, AAS, and RHS. Illustration through cutouts. Simple applications.
 - Problems based on:
 - Angles opposite equal sides are equal and converse.
 - If two sides of a triangle are unequal, then the greater angle is opposite the greater side and converse.
 - Sum of any two sides of a triangle is greater than the third side.
 - Of all straight lines that can be drawn to a given line from a point outside it, the perpendicular is the shortest.
 - Mid-Point Theorem and its converse, equal intercept theorem
 - Pythagoras Theorem
 - Rectilinear Figures
 - Proof and use of theorems on parallelogram.

- Constructions of Polygons
 - Proof and use of Area theorems on parallelograms
- Circle
 - Chord properties
 - Arc and chord properties
- Theme-5: Statistics
 - Introduction, collection of data, presentation of data, Graphical representation of data, Mean, Median of ungrouped data
- Theme-6: Mensuration
 - Area and perimeter of a triangle
 - Area and circumference of circle
 - Surface area and volume of Cube and Cuboids
- Theme-7: Trigonometry
 - Trigonometric Ratios
 - Trigonometric ratios of standard angles
 - Simple 2D problems involving one right-angled triangle
 - Concept of trigonometric ratios of complementary angles and their direct application
- Theme-8: Co-ordinate Geometry
 - Cartesian System
 - Plotting of points in the plane for given coordinates
 - Solving simultaneous linear equations in 2 variables graphically
 - Finding the distance between two points using distance formula